



East West University

Project Report

Semester: Spring-2024

Course Title: Electric Circuit **Course Code:** CSE209 **Sec:** 09

Topic Name: Traffic Signaling System

Group No: 04

Group Members:

Student Name	Student ID
Miftahul Islam Tashfin	2022-3-60-221
Fahim Ahamed Sifat	2022-3-60-034
Aseer Hamim Mahir	2022-3-60-064
Shahriar Rahman	2022-3-60-036

Submitted to-

Rashedul Amin Tuhin

Senior Lecturer

Department of Computer Science and Engineering

East West University

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Problem Statement:

The main objective of this project is to design an efficient traffic signal system for a four-lane intersection. This system aims to smooth traffic flow, reduce congestion, and lower the risk of accidents. In cities, bad traffic signal timing can cause traffic jams, long wait times, higher fuel use, and frustration for drivers and pedestrians. We want to create a system with the best signal timing to help cars and people move smoothly and safely. By adjusting the signal cycle lengths and phase durations based on traffic volume, our system will prevent traffic jams and ensure smooth transitions between signals. This is important for keeping intersections orderly and improving overall traffic flow.

Design:

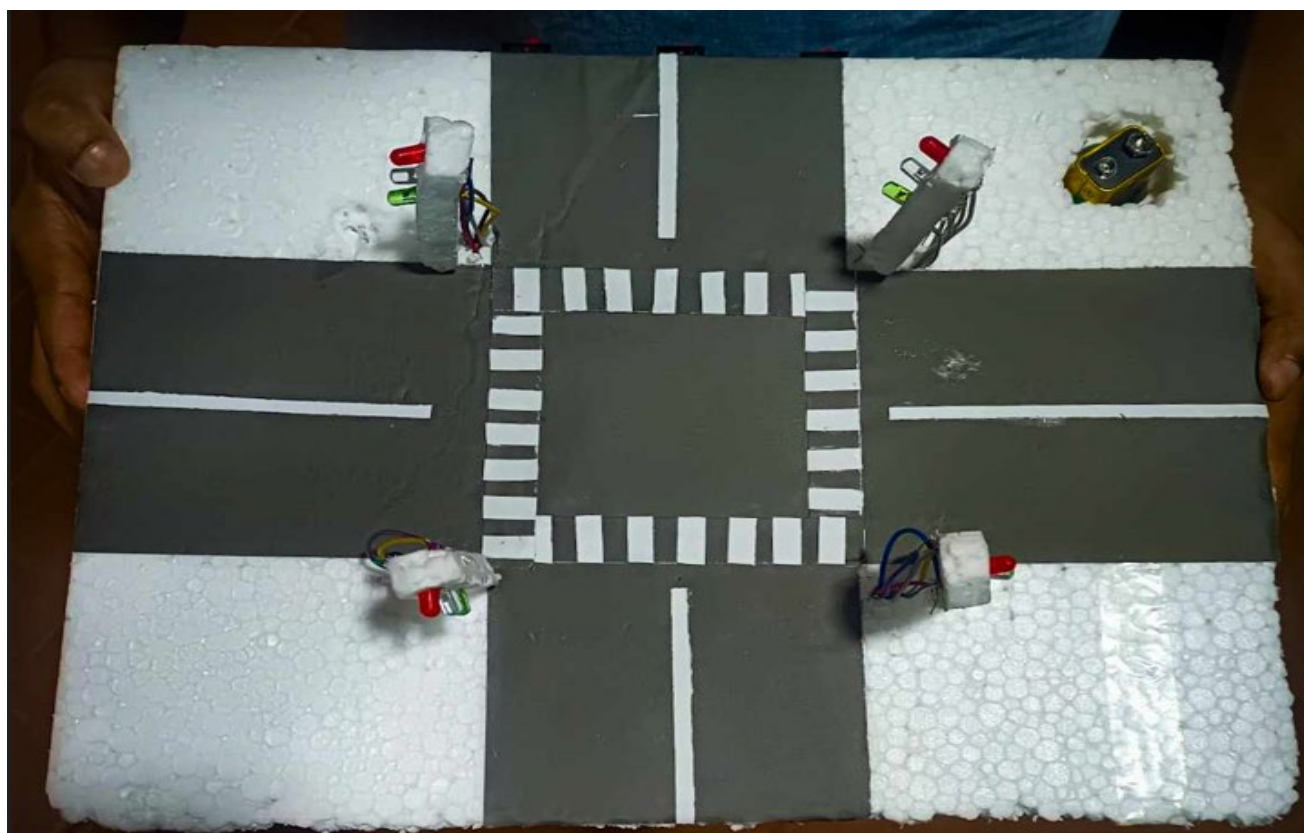
List of the components:

- Breadboard
- Jumper wire
- 1k-ohm resistor (3pcs)
- 9V DC Battery
- Switch (3pcs)
- Red Light (4pcs)
- Yellow Light (4pcs)
- Green Light (4pcs)

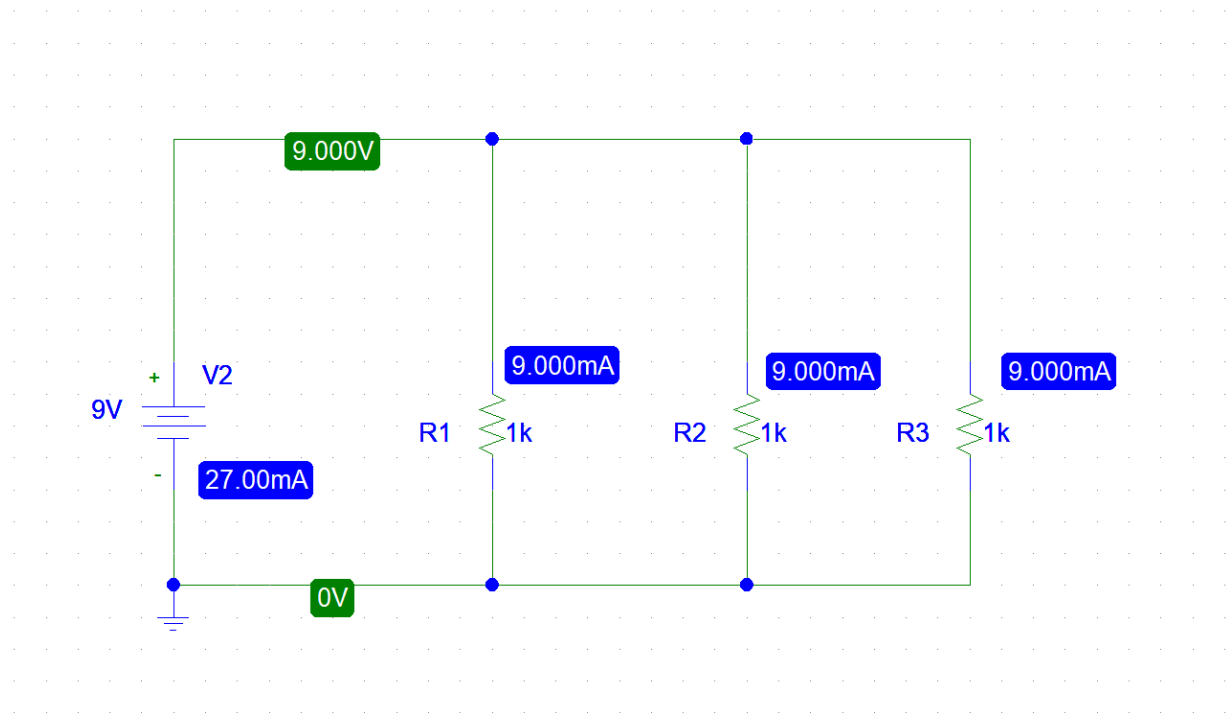
Design Procedure:

In our project, we used a 9V DC Battery and a Breadboard to build the traffic signaling system. The setup included 4 green lights, 4 red lights, and 4 yellow lights, all connected in parallel. Each light had a 1 k-ohm resistor connected in series to limit current and protect the LEDs. The negative terminals of the LEDs were connected to the ground, and the positive terminals were connected to the DC Battery. We connected the negative wire of the DC Battery to the ground of the breadboard to ensure a common ground.

For the signal sequence, the north-south and east-west directions were controlled in the same way, meaning that opposite directions had synchronized signal changes. This setup ensured that traffic flow was managed efficiently, with clear and coordinated signal changes to minimize confusion and improve safety at the intersection.



Simulation and Experimental Results:



The system was tested by manually switching each light on and off to ensure proper functionality. Each set of LEDs (Red, Yellow, Green) responded accurately to the switch positions, confirming the manual control system works as intended for a four-way intersection.

Discussion and Resolved Issues:

What We Learned:

Component Placement: Proper placement of components on a breadboard is crucial for circuit functionality and troubleshooting.

Schematic Reading: Understanding and following schematic diagrams accurately is essential to avoid connection errors and ensure the circuit works as intended.

Resolved Issues:

Component Placement: Carefully placed components on the breadboard to ensure a clear and logical flow of the circuit.

Connection Verification: Double-checked all connections against the circuit diagram to confirm accuracy.

Conclusion:

The Traffic Light Control Project provided an invaluable hands-on learning experience in basic electronics and circuit design. By translating schematic diagrams into a functional circuit, we gained practical skills in component selection, testing, and placement. Implementing debouncing techniques improved the reliability of manual switches, and systematic troubleshooting enhanced our problem-solving abilities. This project not only deepened our understanding of circuit dynamics but also emphasized the importance of teamwork and effective communication in successfully completing technical projects. Overall, the experience has equipped us with foundational skills essential for more advanced electronic projects.